Ex. No: 11

Title: CNN for Image Classification

Packages: TensorFlow, Keras, NumPy, Matplotlib

AIM:

To build and train a Convolutional Neural Network (CNN) for image classification using TensorFlow and Keras.

ALGORITHM:

1. Start the program and import the required libraries.
2. Load the digits images and reshape them to (8, 8, 1).
3. Split the data into training and testing sets.
4. Build the CNN with a Conv2D layer, MaxPooling layer, Flatten layer and Dense layers.
5. Compile the model using the Adam optimizer.
6. Train the model for 15 epochs.
7. Evaluate the model, print the test accuracy and show sample predictions.
8. Plot the training and validation accuracy and stop.

PROGRAM:

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from tensorflow import keras
from tensorflow.keras import layers

# Load digits images and reshape to (8, 8, 1)
X, y = load_digits(return_X_y=True)
X = X.reshape(-1, 8, 8, 1) / 16.0
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)

# CNN model
model = keras.Sequential([
    layers.Input(shape=(8, 8, 1)),
    layers.Conv2D(16, (3, 3), padding="same", activation="relu"),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(32, activation="relu"),
    layers.Dense(10, activation="softmax")
])
model.compile(optimizer="adam", loss="sparse_categorical_crossentropy",
metrics=["accuracy"])
model.summary()

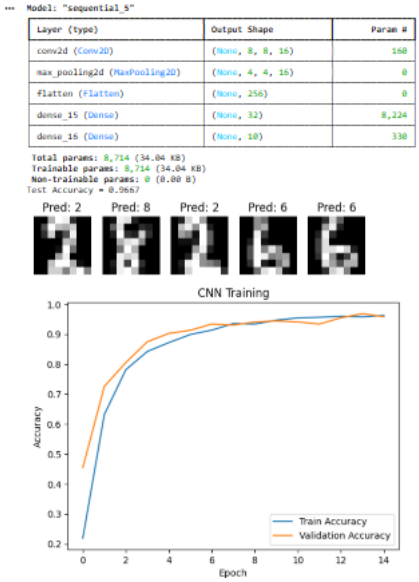
history = model.fit(X_train, y_train, epochs=15, validation_split=0.2, verbose=0)
loss, acc = model.evaluate(X_test, y_test, verbose=0)
print("Test Accuracy =", round(acc, 4))

# Show 5 test predictions
pred = model.predict(X_test[:5], verbose=0).argmax(axis=1)
for i in range(5):
    plt.subplot(1, 5, i + 1)
```

```
    plt.imshow(X_test[i].reshape(8, 8), cmap="gray")
    plt.title("Pred: " + str(pred[i]))
    plt.axis("off")
plt.show()

plt.plot(history.history["accuracy"], label="Train Accuracy")
plt.plot(history.history["val_accuracy"], label="Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("CNN Training")
plt.legend()
plt.show()
```

OUTPUT:



RESULT:

The CNN was built and trained successfully and it classified the images with high test accuracy.